



Koneru Lakshmaiah Education Foundation

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4 Project Abstract Submission Form

Sec No-10	Team No- 16 17	Project Title- Library & Archives Mobile App for Member Self- Services
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1. Abstract

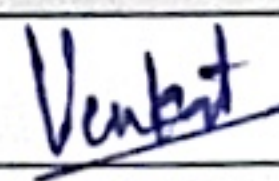
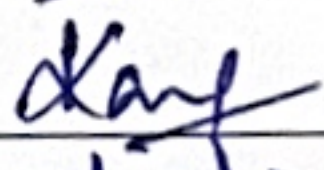
The Library & Archives Mobile App for Member Self-Service is designed to modernize library operations by providing members with convenient access to library resources through a mobile application. In many libraries, users still rely on manual processes or limited web portals to search for books, reserve resources, track due dates, and manage fines. These traditional methods can be time-consuming, less accessible, and often require physical visits to the library.

To address these challenges, the proposed solution offers a mobile self-service platform that enables members to manage library activities anytime and anywhere. The application allows users to register, browse books and assets, reserve available resources, view borrowing history, monitor fines, and receive important notifications. It also supports offline access through local data caching, ensuring that users can access previously synchronized information even without an internet connection.

The system is built using Android (Kotlin) and Jetpack Compose for the mobile interface, with a Node.js REST API and MySQL database for backend operations. Room Database is used for offline storage, while Firebase Cloud Messaging provides real-time notifications for due dates, reservations, and library announcements. Background synchronization ensures consistency between local and server data.

By improving accessibility, reducing manual effort, and enhancing communication between libraries and members, the application provides an efficient, user-friendly, and scalable solution for modern library and archive management.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090034	Venkat Sai	
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3. Frameworks / Platforms / Software Used

- **Android Studio** – Development platform for building and testing the mobile application.
- **Kotlin** – Programming language used for Android app development.
- **Jetpack Compose** – Framework for designing modern and responsive user interfaces.
- **Room Database** – Local database library for offline data storage and caching.
- **Retrofit** – Library for API communication between the mobile app and backend server.
- **Node.js** – Backend runtime environment for developing RESTful services.
- **Express.js** – Framework for creating and managing REST APIs.
- **MySQL** – Database management system for storing and managing application data.
- **Firebase Cloud Messaging (FCM)** – Service for sending push notifications and alerts.
- **Postman** – Tool for API testing and debugging.
- **GitHub** – Platform for version control and project collaboration.

Guide Name and Signature (remarks if any)

